

Serhii Kuzmin

sergeykuzmin495@gmail.com | +1-570-335-0487 | skuzmin.dev | github.com/sersergious | linkedin.com/skuzmin-dev

Education

The University of Scranton, Scranton, PA

May 2026

Bachelor of Science in Computer Science and Mathematical Sciences (Double Major)

GPA: 3.86/4.00

- Dean's List: 8/8 semesters
- Honors Society: Upsilon Pi Epsilon, Pi Mu Epsilon

Technical Skills

Languages: Python, Java, Kotlin, JavaScript/TypeScript, C, SQL, HTML/CSS

Frameworks: FastAPI, Node.js, Bun.js, React.js, Next.js, Three.js, Jetpack Compose

Systems & Infrastructure: Git, Docker, AWS, GitHub/GitLab, GitHub Actions

Databases: PostgreSQL, SQLite, MongoDB

Professional Experience

The University of Scranton, Scranton, PA

October 2023 – December 2025

Peer Tutor

- Provided one-on-one tutoring for Mathematics and Computer Science in Java and C, covering programming fundamentals, object-oriented design, data structures and algorithms, and core low-level programming principles.
- Tutored Mathematics students in Calculus I–III, Linear Algebra, and Discrete Structures, strengthening quantitative reasoning, logical problem solving, and proof-based mathematical thinking.
- Improved students' average academic performance by approximately 30% through customized study plans and targeted skill development.

Projects

Surface Evolver Desktop Application

January 2026 – July 2026

- Ported Surface Evolver from a containerized web app to a cross-platform Tauri desktop app, delivering live frontend viewing/editing with native shell integration at 90% engine compatibility.
- Replaced the FastAPI HTTP backend with a direct FFI sidecar integration, eliminating the network layer between frontend and native engine calls.
- Built a GitHub Actions CI/CD pipeline running the C API test suite, Rust compile checks, and frontend type checks on every deployment.

Recipe Diary Android Application

April 2026 – May 2026

- Designed and implemented a local-first data persistence layer from scratch, using an SQLite schema and Room ORM data access layer to support recipe creation, management, and offline storage.
- Enabled remote data synchronization with a third-party recipe and meal data provider, building a Retrofit-based REST API client to handle network communication between the app and external service.
- Delivered the native application end-to-end using MVVM architecture in Kotlin and Jetpack Compose, with test coverage across unit, integration, and end-to-end layers validating data integrity and API behavior.

Full-Stack Group Chat Application

September 2024 – December 2024

- Engineered a multi-threaded group chat server in C using a thread-per-connection model over POSIX sockets and pthreads, supporting 50+ concurrent users with thread-safe session handling.
- Implemented Argon2id password hashing and a custom binary protocol over WebSockets, reducing message overhead 25% versus a JSON-based baseline.
- Deployed the server on an AWS-hosted FreeBSD instance with Docker for cross-platform compatibility.

Research Experience

The University of Scranton, Scranton, PA

September 2025 – May 2026

Independent Quantum Computing Student Researcher

- Built and verified hybrid quantum simulation experiments in Python using Qiskit, validating results across 8 distinct experiment configurations.
- Quantified the numerical stability of 1- and 2-qubit Z gates in a Variational Quantum Eigensolver implementation by systematically varying readout error in 5% increments.